Myocardial Ischaemia

This guideline is for adults with suspected myocardial ischaemia. If STEMI is present, treat using the relevant STEMI guideline, which may be **'STEMI: primary PCI pathway'**, or **'STEMI: fibrinolytic pathway'**.

- Acquire a 12 lead ECG.
  - Acquire an additional 12 lead ECG using a V4R lead position if an inferior STEMI is suspected.
  - Acquire an additional 12 lead ECG using V7-9 posterior lead positions if a posterior STEMI is suspected.
- Administer 300 mg of aspirin PO.
- Administer 0.4 mg of GTN SL.
- Repeat GTN SL every five minutes provided it relieves symptoms. Increase the dosing interval to 10 minutes if a caution is present.
- Administer an opiate IV if there is significant pain despite GTN.
- Acquire additional 12 lead ECGs every 15-30 minutes looking for evolving changes.

Backup

- EMTs should request backup if the patient has significant pain despite GTN, noting that a Paramedic is appropriate for most patients.
- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

Referral and transport

- All patients with suspected myocardial ischaemia should receive a clear recommendation to be transported to a medical facility by ambulance.
- Transport should usually be to an ED but may be to a primary care facility if this is specifically equipped to investigate patients with suspected myocardial ischaemia.

Additional information

Myocardial Ischaemia

General

- Chest pain/discomfort or shortness of breath without an obvious noncardiac cause in adult patients must be considered to be possibly myocardial ischaemia until proven otherwise. The distribution of the autonomic nerve supply to the intrathoracic and upper abdominal organs is such that the pain/discomfort from myocardial ischaemia may mimic the pain from many other causes in terms of location, sensation and radiation.
- Even if myocardial ischaemia is considered unlikely, a patient with chest pain/discomfort will usually require assessment in an ED because of the possibility of a potentially life-threatening cause, for example, pulmonary embolism, myocarditis, oesophageal tear/rupture, aortic dissection and pneumothorax.
- Personnel must have a very low threshold for clearly recommending that a patient with chest pain/discomfort or shortness of breath without an obvious cause is transported to an ED by ambulance.
- A 12 lead ECG should be obtained in all patients with either typical or atypical symptoms, noting that a normal 12 lead ECG does not rule out myocardial ischaemia. Up to 50% of patients having an acute myocardial infarction have a 12 lead ECG that is initially normal. In the absence of a clear diagnosis, a 12 lead ECG should be repeated every 15-30 minutes looking for evolving ECG changes.
- It is possible for young patients to have myocardial ischaemia. The cause is not usually atherosclerosis of the coronary arteries, but may be from other conditions such as coronary artery spasm, coronary artery dissection or aortic dissection. If a young patient has symptoms suggesting myocardial ischaemia the patient must be given a clear recommendation to be transported to a medical facility by ambulance.

Atypical symptoms

- Some patients have atypical pain or discomfort, including any combination of face, jaw, neck, arm or upper abdominal discomfort.
- Treat as myocardial ischaemia if you suspect atypical symptoms are due to this.

Silent myocardial ischaemia

- Some patients have silent myocardial ischaemia without pain or discomfort.
- Patients who are elderly or have diabetes are particularly at risk of this because they may have autonomic neuropathy.
- Symptoms may include shortness of breath, fatigue, weakness, non-specific malaise or feeling light-headed.

Myocardial Ischaemia

- To make a provisional diagnosis of silent myocardial ischemia there must be signs of myocardial ischaemia on the 12 lead ECG. These signs include ST depression, T wave inversion and T wave flattening.
- The predominant indication for GTN is the presence of chest pain or discomfort, but GTN may be administered if the patient has silent myocardial ischaemia. Only administer repeat doses of GTN if it is clearly associated with improvement of the signs of ischaemia on the 12 lead ECG.

Response and/or transport by helicopter

- Patients with myocardial ischaemia and not STEMI, should usually be transported by road.
- Transport by helicopter may be indicated if the patient has ongoing significant myocardial ischaemia despite treatment and it will save more than 60 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 120 minutes.

More Information

Version history

11 September 2023 1.0.4
Update: Multiple changes.

23 February 2023 1.0.2.1
Update: GTN cautions updated to align with GTN guideline.